

MA 578: Bayesian Statistics

Chapter 9: Model Checking, Comparison, and Selection

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Model Checking

Steps in a Bayesian analysis:

1. Choose a *probabilistic* model.

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What aspects are we checking?

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- ▶ Sampling model
- ▶ Hierarchical structure
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Main question:

Is the observed data “typical” of what we would *expect* if our model were correct?

Posterior Predictive Check (PPC)

Basic idea: simulate *replicated data* from the **posterior predictive distribution**

$$p(y^{\text{rep}} | y) = \int p(y^{\text{rep}} | \theta) p(\theta | y) d\theta,$$

and compare these replications to the observed data.

Discrepancy perspective: We look at the discrepancies between the observed and the replicated data

$$\begin{cases} \text{discrepancies small} & \Rightarrow \text{model fit is acceptable} \\ \text{discrepancies large} & \Rightarrow \text{model is inadequate (revise)} \end{cases}$$

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How to simulate in practice?

We use a Monte Carlo procedure.

Posterior Predictive Check (PPC)

- θ : parameters
- y : observed data
- y^{rep} : replicated data

Posterior predictive distribution of y^{rep} :

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 - ▶ First approximate $p(y^{\text{rep}} | y)$ for each value y^{rep} ;
 - ▶ Then generate samples from the distribution $p(y^{\text{rep}} | y)$.
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 - ▶ $p(y^{\text{rep}} | y)$ is the posterior expectation of $p(y^{\text{rep}} | \theta)$
 - ▶ We can sample $\theta^{(1)}, \dots, \theta^{(S)}$ iid from $p(\theta | y)$ and then approximate $p(y^{\text{rep}} | y)$ with $\frac{1}{S} \sum_{s=1}^S p(y^{\text{rep}} | \theta^{(s)})$
- Feasible only when y^{rep} has small discrete support (or very low dimension)

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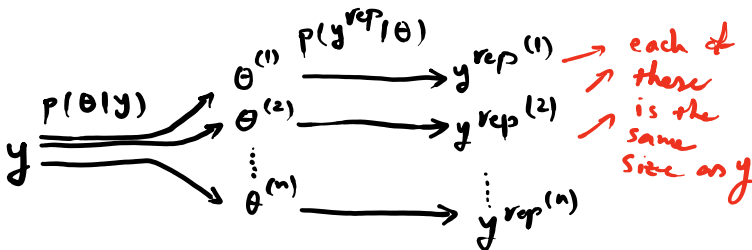
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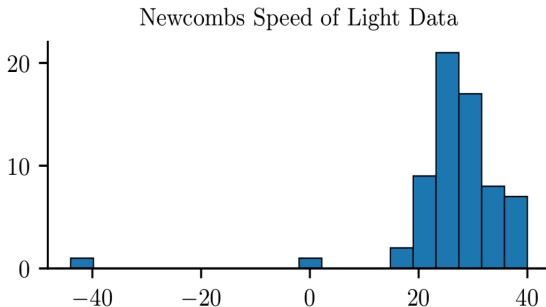


- The sequence $\{(\theta^{(s)}, y_{\text{rep}}^{(s)})\}_{s=1}^S$ are draws from the joint posterior distribution of (θ, y^{rep}) given y
- The sequence $\{y_{\text{rep}}^{(s)}\}_{s=1}^S$ are draws from the *marginal* posterior distribution of y^{rep} , which is the posterior predictive distribution $p(y^{\text{rep}} | y)$.

We use these draws to compute and compare discrepancy measures.

Example: Newcombs Speed-of-Light Data

- In 1882, Newcomb measured the time for light to travel 7,442 meters and obtained $n = 66$ readings.
- The times are recorded as deviations (in nanoseconds) from 24,800 ns.



- There are clear outliers in the **left tail**, suggesting heavy-tailed errors. We will nonetheless fit a **Normal** model as a baseline.
 - ▶ Likelihood: $y_i | \mu, \sigma \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \quad i = 1, \dots, 66.$
 - ▶ Prior (improper noninformative): $p(\mu, \log \sigma) \propto 1$ (Jeffreys prior)

Example: Newcombs Speed-of-Light Data

Goal: simulate replicated datasets y^{rep} of the same size $n = 66$ and compare to the observed data.

Monte Carlo procedure:

1. For $s = 1, \dots, S$: draw $(\mu^{(s)}, \sigma^{(s)}) \sim p(\mu, \sigma \mid y)$.
2. Generate a replicate dataset

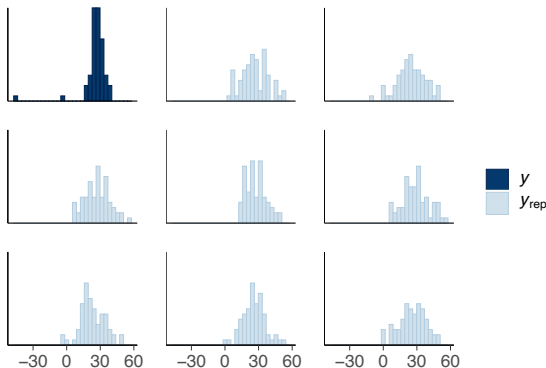
$$y^{\text{rep}(s)} = (y_1^{\text{rep}(s)}, \dots, y_{66}^{\text{rep}(s)}), \quad y_i^{\text{rep}(s)} \stackrel{\text{ind.}}{\sim} \mathcal{N}(\mu^{(s)}, (\sigma^{(s)})^2).$$

3. Repeat for $s = 1, \dots, S$,
4. Compare summaries/plots of $y^{\text{rep}(s)}$ to those of y .

What to look for: tail behavior (extremes), spread, skewness, residual patterns, and other problem-specific features.

PPC: Histogram Overlay

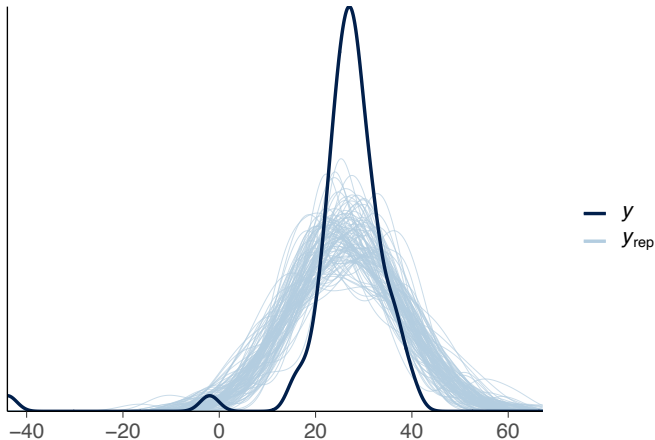
- Compare the observed histogram to several replicated histograms.
- `ppc_hist(y, yrep[1:8, ])` in `bayesplot` package



- None of the replicated histograms show the extreme negative observations present in the real data.
- This indicates the Normal model underestimates the heaviness of the **left** tail.

PPC: Density Overlay

- Overlay observed and replicated densities to assess tail fit.
- `ppc_dens_overlay(y, yrep[1:100, ])` in `bayesplot` package



Measuring Discrepancy

We assess the **discrepancy** between the observed data and their *posterior predictive replications*.

$$\begin{cases} \text{discrepancy small} & \Rightarrow \text{model fit is acceptable} \\ \text{discrepancy large} & \Rightarrow \text{model is inadequate (revise)} \end{cases}$$

How do we measure the **discrepancy**?

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How do we measure the **discrepancy**?

Posterior predictive p -values (PPP-values; Bayesian p -values)

Test quantity (discrepancy function): $T(y, \theta)$

- In contrast to a frequentist *test statistic* $T(y)$, a Bayesian test quantity may depend on the model parameters θ under their posterior distribution.
- Examples: $T(y) = \min_i y_i$, $T(y) = \max_i y_i$, or $T(y, \theta) = \max_i |r_i(\theta)|$ with standardized residuals $r_i(\theta)$.

Measuring Discrepancy

- **Classical (frequentist) p -value** for a test statistic $T(y)$ at parameter value θ (e.g., under H_0):

$$p_F(\theta) = \Pr(T(Y^{\text{rep}}) \geq T(y) \mid \theta),$$

where probability is over a new sample Y^{rep} with θ fixed.

- **Posterior predictive p -value:**

$$p_B = \Pr(T(Y^{\text{rep}}, \Theta) \geq T(y, \Theta) \mid y),$$

where probability is over the joint posterior predictive $p(\Theta, Y^{\text{rep}} \mid y)$.

Equivalent integral form:

$$p_B = \iint \mathbb{I}\{T(y^{\text{rep}}, \theta) \geq T(y, \theta)\} p(y^{\text{rep}} \mid \theta) p(\theta \mid y) dy^{\text{rep}} d\theta.$$

Monte Carlo estimate:

$$\hat{p}_B = \frac{1}{S} \sum_{s=1}^S \mathbb{I}\{T(y_{\text{rep}}^{(s)}, \theta^{(s)}) \geq T(y, \theta^{(s)})\},$$

with $\theta^{(s)} \sim p(\theta \mid y)$ and $y_{\text{rep}}^{(s)} \sim p(y \mid \theta^{(s)})$.

Measuring Discrepancy

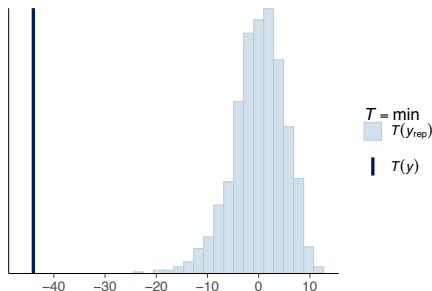
Choose informative test quantities T to target likely failure modes:

Common choices (examples):

- Location/scale: $T(y) = \bar{y}$, $T(y) = \text{Var}(y)$
- Tails/outliers: $T(y) = \max_i y_i$, $T(y) = \#\{i : y_i > c\}$
- Shape: $T(y) = \text{skewness}(y)$
- Residual patterns (model-based): $T(y, \theta) = \sum_i r_i(\theta)^2$, $T(y, \theta) = \max_i |r_i(\theta)|$
(e.g., standardized residuals $r_i(\theta) = \{y_i - \mu_i(\theta)\}/\sigma_i(\theta)$)
- Hierarchical structure: between-group variability, shrinkage amounts, etc.

Measuring Discrepancy

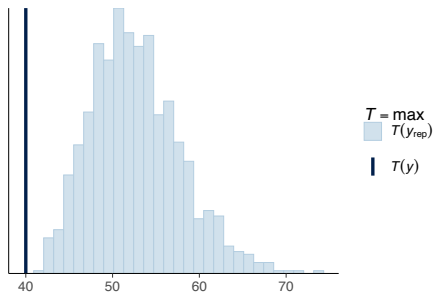
- Newcomb example: left-tail check via min
- Choose $T(y) = \min_i y_i$ to target heavy left tails.
- `bayesplot::ppc_stat(y, yrep, stat = "min")`



- Replicates from the Normal model rarely produce such extreme minima as in the observed data $\Rightarrow$ evidence of lack of fit (left tail).

Measuring Discrepancy

- Newcomb example: right-tail check via max
- Choose $T(y) = \max_i y_i$ to examine the right tail.
- `bayesplot::ppc_stat(y, yrep, stat = "max")`

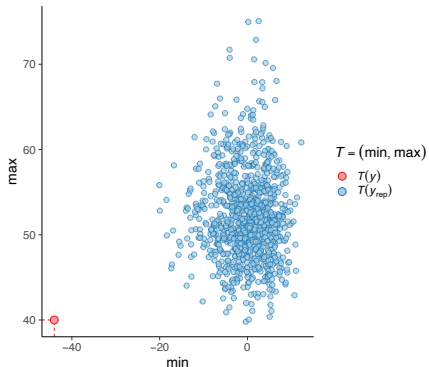


- If the misfit is primarily on the left, max may be less sensitive; still, checking both tails helps diagnose asymmetry.
- In this case, test quantity max also can capture the unfit of the normal model.

Measuring Discrepancy

Bivariate PPC for two discrepancies:

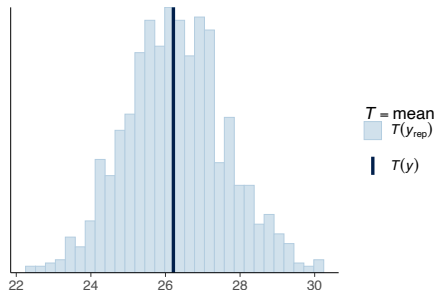
```
bayesplot::ppc_stat_2d(y, yrep, stat = c("min", "max"))
```



- Jointly comparing $(\min y, \max y)$ locates where the observed extremes fall relative to replicated data.

Measuring Discrepancy

- Mean is not sensitive to tail misfit:
- Using $T(y) = \text{mean}(y)$ (the default in `ppc_stat`) often *fails* to detect tail problems.
- `bayesplot::ppc_stat(y, yrep)`



- An incorrect model can still yield accurate inference for *some* aspects of the true population (e.g. the mean).
- Our model should at least generate predictive datasets that resemble the observed dataset in terms of features that are of interest.

Measuring Discrepancy

A test quantity depending on both data and parameter. An example of test quantity (which depends on both θ and y) targeting asymmetry around the location:

$$T(y, \mu) = |y_{(61)} - \mu| - |y_{(6)} - \mu|,$$

where $y_{(k)}$ is the k th order statistic ($n = 66$ here).

- Under a symmetric, well-fit Normal model, $T(y, \mu)$ should concentrate near 0.
- With a heavy *left* tail, $|y_{(6)} - \mu|$ tends to be larger, so $T(y, \mu)$ skews negative.

Model Comparison

- So far: one model at a time (check fit).
- Now: compare two or more models (**binary testing** as a special case).
- Two complementary approaches:
 - ▶ *Predictive comparison*: We compare their expected predictive accuracy (e.g., expected out-of-sample accuracy).
 - ▶ *Bayesian model choice*: assign each model a prior probability and update via Bayes rule.

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 - ▶ *Predictive comparison*: We compare their expected predictive accuracy (e.g., expected out-of-sample accuracy).
 - ▶ *Bayesian model choice*: assign each model a prior probability and update via Bayes rule.
- Posterior model probability is proportional to **prior** $\times$ **marginal likelihood**:

$$p(H_k | y) \propto p(y | H_k) p(H_k), \quad p(y | H_k) = \int p(y | \theta_k, H_k) p(\theta_k | H_k) d\theta_k.$$

Posterior Odds and Bayes Factors

- Hypotheses:

$$\begin{cases} H_0 : \text{model 0 with prior } p(H_0) \\ H_1 : \text{model 1 with prior } p(H_1) \end{cases}$$

- We can make a decision if we can compute the posterior odds:

$$\frac{p(H_1 | y)}{p(H_0 | y)}$$

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$$\frac{p(H_1 | y)}{p(H_0 | y)} = \underbrace{\frac{p(y | H_1)}{p(y | H_0)}}_{\text{Bayes factor } BF(H_1; H_0)} \times \underbrace{\frac{p(H_1)}{p(H_0)}}_{\text{prior odds}}.$$

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- Therefore, if we can compute the Bayes factor we will be able to decide which hypothesis to choose.
- Bayes factor:

$$BF(H_1; H_0) = \frac{p(y | H_1)}{p(y | H_0)} = \frac{\int p(y | \theta_1, H_1) p(\theta_1 | H_1) d\theta_1}{\int p(y | \theta_0, H_0) p(\theta_0 | H_0) d\theta_0}.$$

- $BF > 1$ favors H_1 .
- Note that the Bayes factor does not depend on the prior for hypotheses, but does depend on the prior for the parameter of interest in each hypothesis

Posterior Odds and Bayes Factors

- Posterior distribution:

$$\begin{aligned} p(H_0 | y) &= \frac{p(y | H_0)p(H_0)}{p(y | H_0)p(H_0) + p(y | H_1)p(H_1)} \\ &= \frac{1}{1 + \frac{p(y|H_1)p(H_1)}{p(y|H_0)p(H_0)}} \\ &= \frac{1}{1 + BF(H_1; H_0) \times \text{Prior odds}}. \end{aligned}$$

Example: The Sally Clark Case

- Starting in 1996, four British mothers were arrested for murder because two or more of their infants had died in their cribs. Three of the mothers were convicted and sentenced to life imprisonment.
- Sally Clark, a lawyer who lost her first son at 11 weeks and her second at 8 weeks, was convicted in 1999. A prominent pediatrician, Sir Roy Meadow, had testified for the prosecution about Sudden Infant Death Syndrome, known as SIDS. Meadow said the incidence of one SIDS death was one in 1300. Then, Meadow assumed erroneously that each sibling's death occurred independently of the other. Multiplying 1300 by 1300, he calculated that the chance of two children dying in a family was so rare—one in 1.7 million—that they must have been murdered.
- A Bayesian analysis would have shown that the children most probably died of SIDS.

Example: The Sally Clark Case

- According to Bayes' rule, a highly unlikely event can happen but it must be compared with other highly unlikely events. Thus, the question before the court should have been:

Did the Clark babies more likely die of natural causes or murder?

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- Natural causes of sudden infant death:
 $p(\text{one random infant dying from SIDS}) = 1/1300$
The estimated odds of a second SIDS death in the same family was much larger: one in 100, because family members can share a common environmental or genetic propensity for SIDS.
- The hypothesis that the babies were murdered:
Only about 30 children out of 650,000 annual births in England, Scotland, and Wales were known to have been murdered by their mothers.
The number of double murders must be much lower, estimated as 10 times less likely.

- $\begin{cases} H_0 : \text{Sally Clark's two children died of SIDS} \\ H_1 : \text{Sally Clark's two children did not die of SIDS} \end{cases}$
Data y : both children died suddenly and unexpectedly.
- Priors:

$$p(H_0) = 1/1300 \times 1/100 = 0.0000077$$

$$p(H_1) = 1 - p(H_0) = 0.9999923$$

- Bayes factor:

$$p(y | H_0) = 1$$

$$p(y | H_1) \approx p(\text{Sally Clark's two children were murdered}) = 30/650000 \times 1/10 = 0.0000046$$

- Posterior distribution:

$$p(H_0 | y) = \frac{1}{1 + BF(H_1; H_0) \times \text{Prior odds}} = \frac{1}{1 + \frac{0.0000046}{1} \times \frac{0.9999923}{0.0000077}} \approx 0.63$$

- Thus, it is more than likely that the infants died of SIDS.

Lindley's Paradox

- In a city, $y = 49,581$ boys and $n - y = 48,870$ girls were born ($n = 98,451$).
- Model: $Y \sim \text{Binomial}(n, \theta)$
- Proportion of male births (Sample proportion $\hat{\theta}$): $\frac{y}{n} = \frac{49581}{98451} \approx 0.5036$.
- We are interested in testing

$$\begin{cases} H_0 : \theta = 0.5, \\ H_1 : \theta \neq 0.5. \end{cases}$$

Frequentist Analysis (large- n Normal Approximation)

- Under H_0 , $Y \approx \mathcal{N}(\mu, \sigma^2)$ with

$$\mu = n \cdot 0.5 = 49,225.5, \quad \sigma^2 = n \cdot 0.5 \cdot 0.5 = 24,612.75.$$

- Two-sided p -value:

$$p\text{-val} = 2 \Pr\left(Z \geq \frac{y - \mu}{\sigma}\right) = 2 \Pr(Z \geq 2.27) \approx 0.0235.$$

- Conclusion: the frequentist approach rejects H_0 at 5%.

Bayesian Analysis

- We consider the following non-informative priors: $p(H_0) = p(H_1) = \frac{1}{2}$
 $p(\theta | H_1) = \text{Unif}(0, 1) = \text{Beta}(1, 1)$
- Bayes factor

$$\begin{aligned}BF(H_1; H_0) &= \frac{p(y | H_1)}{p(y | H_0)} \\&= \frac{\int p(y | \theta, H_1)p(\theta | H_1)}{\int p(y | \theta, H_0)p(\theta | H_0)} \\&= \frac{\int \binom{98451}{49581} (\theta)^{49581} (1 - \theta)^{98451 - 49581} p(\theta | H_1) d\theta}{\binom{98451}{49581} (0.5)^{49581} (1 - 0.5)^{98451 - 49581}} \\&= \frac{\binom{98451}{49581} \frac{\Gamma(49581+1)\Gamma(98451-49581+1)}{\Gamma(98451+2)}}{\binom{98451}{49581} (0.5)^{49581} (1 - 0.5)^{98451 - 49581}} \\&= 0.052.\end{aligned}$$

- Posterior distribution

$$p(H_0 | y) = \frac{1}{1 + BF(H_1; H_0) \times \text{Prior odds}} = \frac{1}{1 + 0.0523 \times 1} \approx 0.95$$

- Therefore, the Bayesian approach strongly favors H_0 .

Bayesian Analysis

Also note that with a prior distribution $p(\theta) \sim \text{Beta}(1, 1)$, we have the following posterior distribution

$$p(\theta | y) = \text{Beta}(49581 + 1, 98451 - 49581 + 1)$$

Therefore,

$$p(\text{more male births} | y) = \int_{0.5}^1 p(\theta | y) d\theta = 0.983$$

Reconciling the “Paradox”

- The frequentist approach rejects H_0
- The Bayesian approach strongly favors H_0
- Our posterior belief indicates that it is very likely that the proportion of male births is above 0.5.

What is going on?!

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- Our posterior belief indicates that it is very likely that the proportion of male births is above 0.5.

What is going on?!

- Frequentist test ignores $p(y \mid H_1)$ and asks if y is unusual under H_0 .
- The Bayesian approach is answering a different question!
It decides between H_0 and a combination of all other possibilities. Majority of those other possibilities are highly unlikely. Hence, the Bayesian approach finds H_0 more favorable.
- If we set a *focused* alternative, e.g. $H_1 : \theta = 0.504$, the Bayesian approach can agree with the frequentist test.

Final remark: Lindleys paradox highlights the different questions answered by tests vs. model comparison and the sensitivity of Bayes factors to priors under H_1 .

Comparing K Hypotheses via Decision Theory

- Let $\ell(a, H_k)$ be the loss if we take action $a = \hat{h}(y)$ (i.e., choosing hypothesis $\hat{h}(y)$) when the true hypothesis is H_k .
- Posterior expected loss of action a :

$$\mathbb{E}[\ell(a, H^*) \mid y] = \sum_{k=1}^K \ell(a, H_k) p(H_k \mid y).$$

- **Optimal decision** minimizes posterior expected loss.

Comparing K Hypotheses via Decision Theory

- **Example (0-1 loss):** $\ell(a, H_k) = \mathbf{1}\{a \neq H_k\}$. Then for action $a = H_j$,

$$\mathbb{E}[\ell(H_j, H^*) \mid y] = 1 - p(H_j \mid y).$$

Hence the optimal choice is

$$\hat{h}^*(y) = \arg \max_k p(H_k \mid y),$$

- i.e., pick the hypothesis with the maximum posterior probability.
- This is Maximum a posteriori (MAP) estimation.

Pairwise Bayes Factors

- Assign prior probabilities over models $p(M_k)$.
- Marginal likelihood (model evidence):

$$p(y | M_k) = \int p(y | \theta_k, M_k) p(\theta_k | M_k) d\theta_k.$$

- Posterior model probabilities:

$$p(M_k | y) = \frac{p(M_k) p(y | M_k)}{\sum_{\ell=1}^K p(M_\ell) p(y | M_\ell)}.$$

- Bayes factor comparing M_a to M_b :

$$BF_{ab} = \frac{p(y | M_a)}{p(y | M_b)}.$$

- Posterior odds factor into prior odds $\times$ Bayes factor:

$$\frac{p(M_a | y)}{p(M_b | y)} = \frac{p(M_a)}{p(M_b)} \times BF_{ab}.$$

Pairwise Bayes Factors

- Consider the following rule: choose M_k iff

$$BF_{ki} \geq 1 \quad \text{for all } i.$$

- How is this rule related to MAP estimator?

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- How is this rule related to MAP estimator?
- This rule coincides with the MAP model when the model priors are uniform.

Model Selection in Regression Analysis

In many real-world applications,

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- we believe the majority have no true association with the outcome;

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- we believe the majority have no true association with the outcome;
- including all candidates often leads to poor estimation and prediction.
⇒ Include only variables with substantial evidence of association.

Model Selection in Regression Analysis

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- we believe the majority have no true association with the outcome;
- including all candidates often leads to poor estimation and prediction.
⇒ Include only variables with substantial evidence of association.
- This is the **variable selection** problem, a special case of **model selection**.

Imposing Sparsity via Spike-and-Slab

- Encode sparsity through a prior that gives *positive probability* to $\beta_j = 0$.
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Imposing Sparsity via Spike-and-Slab

- Encode sparsity through a prior that gives *positive probability* to $\beta_j = 0$.
- A Normal prior does not do this (it has density, not mass, at zero).
- **Indicator parameterization:** write $\beta_j = z_j b_j$ with $z_j \in \{0, 1\}$ and $b_j \in \mathbb{R}$.

Model (with Gaussian noise):

$$y_i = \sum_{j=1}^p z_j b_j x_{ij} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).$$

- z_j indicates inclusion of predictor j ; each $z = (z_1, \dots, z_p)$ defines a model.
- Bayesian model selection proceeds by *posterior inference on z* .

Posterior over Models

- Posterior over models:

$$p(z \mid y, X) = \frac{p(z) p(y \mid X, z)}{\sum_{z'} p(z') p(y \mid X, z')}.$$

- With conjugate priors, $p(y \mid X, z)$ is available in closed form.

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- With conjugate priors, $p(y \mid X, z)$ is available in closed form.
- Note that having $p(y \mid X, z)$ also enables use to compare any two models z_a vs z_b via **posterior odds**:

$$\underbrace{\frac{p(z_a \mid y, X)}{p(z_b \mid y, X)}}_{\text{posterior odds}} = \underbrace{\frac{p(z_a)}{p(z_b)}}_{\text{prior odds}} \times \underbrace{\frac{p(y \mid X, z_a)}{p(y \mid X, z_b)}}_{\text{Bayes factor } BF(z_a; z_b)}.$$

Posterior over Models

- Prior on models: $p(z)$

The most common choice: uniform prior:

$$z_i \stackrel{iid}{\sim} \text{Bernoulli}(\frac{1}{2})$$

- Prior on parameters for each model:

$$\sigma^2 \sim \text{Inv-Gamma}(\nu_0/2, \nu_0\sigma_0^2/2)$$

$$\beta_z \mid X, z, \sigma^2 \sim \mathcal{N}(0, g\sigma^2[X_z^\top X_z]^{-1})$$

- where, defining p_z as the number of non-zero entries of z ,
 - ▶ X_z : $n \times p_z$ matrix corresponding to variables j for which $z_j = 1$
 - ▶ β_z : $p_z \times 1$ vector consisting of entries of β for which $z_j = 1$
 - ▶ $\beta_j = 0$ for j 's s.t. $z_j = 0$
- Given z , the prior on β_z is a g -prior.

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- Given z , the prior on β_z is a g -prior.

- Under this model, the marginal will be

$$p(y \mid X, z) = \pi^{-n/2} \frac{\Gamma([\nu_0 + n]/2)}{\Gamma(\nu_0/2)} (1 + g)^{-p_z/2} \frac{(\nu_0 \sigma_0^2)^{\nu_0/2}}{(\nu_0 \sigma_0^2 + SSR_g^z)^{(\nu_0 + n)/2}}$$

$$\text{where, } SSR_g^z = y^\top (I - \frac{g}{g+1} X_z (X_z^\top X_z)^{-1} X_z^\top) y$$

Gibbs Sampling

- Having $p(y \mid X, z)$, we can conceptually pick the model which maximizes

$$p(z \mid y, X) \propto p(y \mid X, z)p(z).$$

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However, this is not practical (why?)

- If we allow each of the p regression coefficients to be either zero or non-zero, then there are 2^p different model to consider
- For instance, in the example that we will see later with $p = 64$, the total number of models is $2^{64} = 1.8 \times 10^{19}$
- It is not feasible to compute $p(y | X, z)$ (and hence $p(z | y, X)$) for all possible z .
- However, we can approximate the posterior $p(z | y, X)$ via Gibbs sampling.

Gibbs Sampling

- Define

$$o_j = \frac{p(z_j = 1 \mid X, y, z_{-j})}{p(z_j = 0 \mid X, y, z_{-j})} = \frac{p(y \mid X, z_{-j}, z_j = 1)}{p(y \mid X, z_{-j}, z_j = 0)} \times \frac{p(z_{-j}, z_j = 1)}{p(z_{-j}, z_j = 0)}$$

- Note that $\frac{p}{1-p} = o_j \Rightarrow p = \frac{o_j}{1+o_j}$
- Therefore,

$$p(z_j = 1 \mid X, y, z_{-j}) = \frac{o_j}{1 + o_j}$$

- Therefore, we can update z_j 's one by one

Gibbs Sampling

- After obtaining a new sample $z^{(s)}$ by updating $z_1, \dots, z_p$, we can sample (β, σ^2) directly from their posterior (given model $z^{(s)}$) using the results from Chapter 6

$$\begin{array}{ccccc} z^{(s)} & \longrightarrow & \sigma^{2(s)} & \longrightarrow & \beta^{(s)} \\ \downarrow & & & & \\ z^{(s+1)} & \longrightarrow & \sigma^{2(s+1)} & \longrightarrow & \beta^{(s+1)} \end{array}$$

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- Sampling (β, σ^2) given a model:**
- Given z , the conjugate posteriors are:

$$\sigma^2 \mid y, X, z \sim \text{Inv-Gamma}\left(\frac{\nu_0 + n}{2}, \frac{\nu_0 \sigma_0^2 + SSR_g^z}{2}\right),$$

$$b_z \mid \sigma^2, y, X, z \sim \mathcal{N}\left(\frac{g}{g+1} \hat{b}_z, \frac{g}{g+1} \sigma^2 [X_z^\top X_z]^{-1}\right), \quad \hat{b}_z = (X_z^\top X_z)^{-1} X_z^\top y.$$

- Sampler outline per iteration s :**

- For $j = 1:p$, update $z_j^{(s)} \sim p(z_j \mid y, X, z_{-j}^{(s)})$.
- Draw $\sigma^{2(s)} \sim p(\sigma^2 \mid y, X, z^{(s)})$ and $b_z^{(s)} \sim p(b_z \mid \sigma^{2(s)}, y, X, z^{(s)})$.

Gibbs Sampling

- Note that z_j is sampled from conditional distribution $p(z_j \mid y, X, z_{-j})$ as opposed to the full conditional distribution $p(z_j \mid y, X, \sigma^2, \beta, z_{-j})$
- This is not a problem, because the distribution of $z^{(s)}$ is converging to the target posterior distribution $p(z \mid y, X)$ and $(\sigma^{2(s)}, \beta^{(s)})$ is sampled from $p(\sigma^2, \beta \mid z^{(s)}, y, X)$
 $\Rightarrow$ The distribution of $(\sigma^{2(s)}, \beta^{(s)})$ is converging to $p(\sigma^2, \beta \mid y, X)$

Decision Rule

- **In principle:** under 0–1 loss on the model index, the Bayes-optimal single model is the *MAP model* (maximum posterior probability model).
- **However**, as mentioned earlier, if we allow each of the p regression coefficients to be either zero or non-zero, then there are 2^p different models to consider.
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 - ▶ It is not feasible to compute $p(y | X, z)$ (and hence $p(z | y, X)$) for all possible z .
- **In practice:** instead of exhaustively searching for the global MAP model, we use MCMC (Gibbs sampling) to explore the posterior over models:
 - ▶ we visit a collection of **relatively high posterior probability** models,
 - ▶ we approximate **posterior inclusion probabilities (PIP)** for predictors.
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 - ▶ we construct a predictor based on a list of relatively high probability models via **Bayesian model averaging (BMA)**.
 - ▶ MAP, PIP, and BMA can all be considered as special cases of the general Bayesian model comparison approach that we introduced earlier.

Comparing K Hypotheses via Decision Theory

- Suppose we have K competing hypotheses $H_1, \dots, H_K$.
 - ▶ In model selection, each H_k can be “model M_k is the data-generating model”.
- We choose an **action** $a = \hat{h}(y)$ after observing data y .
- Let $\ell(a, H_k)$ be the loss if we take action a when the true hypothesis is H_k .
- Posterior expected loss of action a :

$$\mathbb{E}[\ell(a, H^*) \mid y] = \sum_{k=1}^K \ell(a, H_k) p(H_k \mid y).$$

- **Bayes rule / optimal decision:**

$$\hat{a}^*(y) = \arg \min_a \mathbb{E}[\ell(a, H^*) \mid y].$$

Comparing K Hypotheses via Decision Theory

- Different choices of loss ℓ give different procedures:
 - ▶ 0–1 loss on whole models $\Rightarrow$ MAP model.
 - ▶ Additive 0–1 loss per variable $\Rightarrow$ median probability model (MPM).
 - ▶ Predictive loss (e.g. squared error for y_{new}) $\Rightarrow$ BMA predictor.

MAP Model Selection

- Loss function:

$$\ell(a, H_k) = \mathbf{1}\{a \neq H_k\}.$$

- Consider actions restricted to choose one hypothesis: $a = H_j$. Then

$$\mathbb{E}[\ell(H_j, H^*) \mid y] = \sum_{k=1}^K \mathbf{1}\{H_j \neq H_k\} p(H_k \mid y) = 1 - p(H_j \mid y).$$

- Hence the optimal choice is

$$\hat{h}^*(y) = \arg \min_j \mathbb{E}[\ell(H_j, H^*) \mid y] = \arg \max_k p(H_k \mid y).$$

- This is the **MAP rule** (maximum a posteriori):
 - ▶ In model-selection language, it picks the single model with largest posterior probability.
- As we said earlier, even in moderate size regression problems, number of models is exponentially large and hence we are not able to look at all posteriors.
 - ▶ Therefore, MAP model selection may not be feasible.

Posterior Inclusion Probabilities

- Another approach for Bayesian model selection is based on using “Posterior inclusion probabilities (PIPs)”:

Given the model indicators $z = (z_1, \dots, z_p)$,

- Each $z_j \in \{0, 1\}$ encodes whether predictor j is included.
- The **posterior inclusion probability** (PIP) for predictor j is

$$\pi_j = p(z_j = 1 \mid y, X).$$

- We can estimate π_j from MCMC samples $z^{(1)}, \dots, z^{(S)}$:

$$\hat{\pi}_j = \frac{1}{S} \sum_{s=1}^S \mathbf{1}\{z_j^{(s)} = 1\}.$$

- $\hat{\pi}_j$ summarizes how strongly the data support including predictor j , integrating over all other variables and all models.

Posterior Inclusion Probabilities

- Let $|z| = \sum_{j=1}^p z_j$ be the **model size** (number of included predictors).
- The posterior distribution of model size is

$$p(|z| = k \mid y, X) = \sum_{z: |z|=k} p(z \mid y, X).$$

- From MCMC output, estimate via

$$\hat{p}(|z| = k \mid y, X) = \frac{1}{S} \sum_{s=1}^S \mathbf{1}\{|z^{(s)}| = k\}.$$

- This gives a **global view of sparsity**: do we expect a very small model, a medium-sized model, etc.?
- Useful for checking whether the prior $p(z)$ imposed a reasonable amount of shrinkage.

Posterior Inclusion Probabilities

- Now we want an **action** $a = (a_1, \dots, a_p) \in \{0, 1\}^p$:

$$a_j = \begin{cases} 1 & \text{we include predictor } j \text{ in our final model,} \\ 0 & \text{we exclude predictor } j. \end{cases}$$

- Let z be the true (unknown) inclusion vector.
- A simple **additive loss**:

$$\ell(a, z) = \sum_{j=1}^p [c_{FP} \mathbf{1}\{a_j = 1, z_j = 0\} + c_{FN} \mathbf{1}\{a_j = 0, z_j = 1\}],$$

where

- ▶ c_{FP} : cost of a false positive (including a noise variable),
- ▶ c_{FN} : cost of a false negative (excluding a truly relevant variable).
- Goal: choose a that minimizes posterior expected loss

$$\mathbb{E}[\ell(a, Z) \mid y, X].$$

Posterior Inclusion Probabilities

Because the loss is additive over j :

$$\mathbb{E}[\ell(\mathbf{a}, Z) \mid y, X] = \sum_{j=1}^p \mathbb{E}[\ell_j(a_j, Z_j) \mid y, X],$$

we can optimize each a_j separately.

For a single predictor j :

- Expected loss if we set $a_j = 1$:

$$\mathbb{E}[\ell_j(1, Z_j) \mid y, X] = c_{FP} (1 - \pi_j).$$

- Expected loss if we set $a_j = 0$:

$$\mathbb{E}[\ell_j(0, Z_j) \mid y, X] = c_{FN} \pi_j.$$

- Choose $a_j = 1$ if $\mathbb{E}[\ell_j(1, Z_j) \mid y, X] < \mathbb{E}[\ell_j(0, Z_j) \mid y, X]$, i.e.

$$c_{FP} (1 - \pi_j) < c_{FN} \pi_j \iff \pi_j > \frac{c_{FP}}{c_{FP} + c_{FN}}.$$

Median Probability Model

- If we treat false positives and false negatives symmetrically:

$$C_{FP} = C_{FN} \quad \Rightarrow \quad \pi_j > \frac{1}{2} \text{ is the optimal rule.}$$

- Then the optimal action is:

$$a_j^* = \begin{cases} 1 & \text{if } \pi_j = p(z_j = 1 \mid y, X) > 0.5, \\ 0 & \text{if } \pi_j \leq 0.5. \end{cases}$$

- The resulting model is called the **median probability model** (MPM):

$$\mathcal{M}_{\text{MPM}} = \{j : \pi_j > 0.5\}.$$

- MPM is based on applying the MAP rule separately to each 2-hypothesis problem $z_j = 0$ vs $z_j = 1$.
- Under certain conditions (e.g. nearly orthogonal predictors and squared-error prediction), the median probability model is optimal for **prediction among single models** (not average models): it minimizes the posterior expected predictive risk among all models.

Bayesian Model Averaging

- Another approach to use Gibbs samples from the posterior is to construct a model based on a list of relatively high probability models via **Bayesian model averaging (BMA)**.
- For squared-error loss on predictions, the BMA predictor (posterior predictive mean) is always Bayes-optimal.
- We look at this approach through an example.

Example: Diabetes Study

$n = 442$ diabetes patients

y : disease progression over one year

$\mathbf{x}_i \in \mathbb{R}^{64}$: covariates

Goal: To build a predictive regression model.

That is, to find $\hat{\beta}$ so we can predict a new case as $\tilde{\mathbf{x}}^\top \hat{\beta}$

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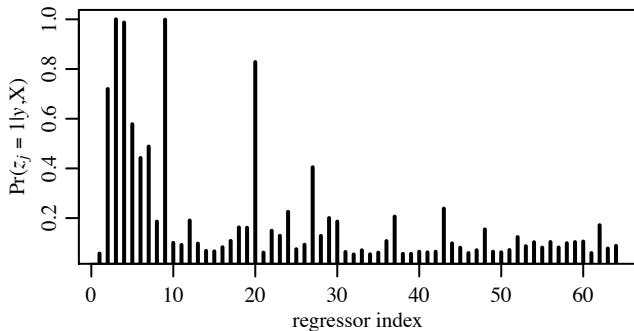
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- We run a Gibbs sampler for $S = 10,000$ iterations
- We get 10^4 samples of z, σ^2, β that we use to approximate the posterior distribution $p(z, \sigma^2, \beta \mid y, X)$
- Recall that the total number of models $= 2^{64} \approx 10^{19}$
- This is 10^{15} times as large as the number of samples we have!
 $\Rightarrow$ Only a small fraction of the models are ever visited by the sampler

- However, if most of regressors have no effect, majority of the models should have negligible probability. So, it is okay that the sampler does not visit them
- Although the joint distribution of z_j 's and β_j 's is not going to be approximated very well, the Gibbs sampler can still provide reasonable estimate of the marginal posterior distribution of individual z_j 's and β_j 's



Posterior probabilities that each coefficient is non-zero.

- We use posterior mean as our estimator:

$$\hat{\beta}_{bma} = \frac{1}{S} \sum_{s=1}^S \beta^{(s)}$$

- This is called the Bayesian model averaged estimate of β

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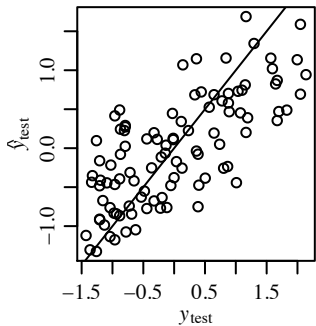
- This is called the Bayesian model averaged estimate of β
- Therefore, note that we ended up with **model averaging** instead of **model selecting**, but we got help from sparsity
- The “model averaging” terminology emphasizes the fact that we use the posterior over all models rather than selecting a single model.

Question: How do we evaluate a predictor?

- Randomly divide the 442 subjects into
 - ▶ 342 training samples (y, X)
 - ▶ 100 test samples (y_{test}, X_{test})
- Fit the regression model to obtain $\hat{\beta}$ using the training data
- Use the estimated regression coefficient $\hat{\beta}$ to generate $\hat{y}_{test} = X_{test}\hat{\beta}$
- Evaluate the performance by comparing y_{test} and $\hat{y}_{test}$, e.g., via MSE (mean squared error):

$$\frac{1}{100} \sum (y_{test_i} - \hat{y}_{test_i})^2$$

- In our case we use model averaged predicted values $\hat{y}_{test} = X_{test}\hat{\beta}_{bma}$



y_{test} versus predictions based on the model averaged estimate of β .
Test MSE = 0.452 for Bayesian variable selection

Optimality of BMA for Point Prediction

- Goal: predict a future response y_{new} at covariate x_{new} .
- Action: choose a point predictor $\hat{y}(x_{\text{new}})$.
- Loss: squared prediction error

$$L(\hat{y}, y_{\text{new}}) = (y_{\text{new}} - \hat{y}(x_{\text{new}}))^2.$$

- Posterior expected loss for a fixed t :

$$\mathbb{E}[(Y_{\text{new}} - t)^2 \mid x_{\text{new}}, y, X] = \text{var}(Y_{\text{new}} \mid x_{\text{new}}, y, X) + (\mathbb{E}[Y_{\text{new}} \mid x_{\text{new}}, y, X] - t)^2,$$

minimized at

$$t^* = \mathbb{E}[Y_{\text{new}} \mid x_{\text{new}}, y, X].$$

- With model uncertainty (models indexed by z),

$$\mathbb{E}[Y_{\text{new}} \mid x_{\text{new}}, y, X] = \sum_z p(z \mid y, X) \mathbb{E}[Y_{\text{new}} \mid x_{\text{new}}, y, X, z],$$

which is exactly the **BMA predictor** $\hat{y}_{\text{BMA}}(x_{\text{new}})$.

⇒ Under squared-error loss, the BMA predictor is the **Bayes-optimal point predictor** for y_{new} .

MPM vs. MAP vs. BMA

- **MAP model:** model z with largest posterior probability $p(z \mid y, X)$.
- **MPM:** includes all predictors with $\pi_j > 0.5$.
- **BMA predictor:** averages over all models,

$$\hat{y}_{\text{BMA}}(x_{\text{new}}) = \mathbb{E}[Y_{\text{new}} \mid x_{\text{new}}, y, X] = \sum_z p(z \mid y, X) \mathbb{E}[Y_{\text{new}} \mid z, x_{\text{new}}, y, X].$$

- These three can give different selected models:
 - ▶ MAP favors the single most probable configuration of all z_j 's.
 - ▶ MPM focuses on **marginal** inclusion for each predictor.
 - ▶ BMA does not select a single model at all, but often gives best predictive accuracy.
- In practice, it is common to report:
 - ▶ the BMA predictor, and
 - ▶ the median probability model as a sparse, interpretable summary.